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Drivers of diversity gradients of a highly mobile marine assemblage in a mesoscale seascape

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ABSTRACT: With the unprecedented rate of biodiversity change in the world today, understanding how diversity gradients are maintained at mesoscales is a key challenge. Drawing on information provided by 3 comprehensive fishery surveys (conducted in different years but in the same season and with the same sampling design), we used boosted regression tree (BRT) models in order to relate spatial patterns of α-diversity in a demersal fish assemblage to environmental variables in the San Matias Gulf (Patagonia, Argentina). We found that, over a 4 yr period, persistent diversity gradients of species richness and probability of an interspecific encounter (PIE) were shaped by 3 main environmental gradients: bottom depth, connectivity with the open ocean, and proximity to a thermal front. The 2 main patterns we observed were: a monotonic increase in PIE with proximity to fronts, which had a stronger effect at greater depths; and an increase in PIE when closer to the open ocean (a 'bay effect' pattern). The originality of this work resides on the identification of high-resolution gradients in local, demersal assemblages driven by static and dynamic environmental gradients in a mesoscale seascape. The maintenance of environmental gradients, specifically those associated with shared resources and connectivity with an open system, may be key to understanding community stability.

KEY WORDS: Demersal fishes · Species–energy relationships · Spatial structure · Marine fronts · α-diversity · Bay effect · Boosted regression trees · Connectivity

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1. INTRODUCTION

Biodiversity is changing at an unprecedented rate (Dornelas et al. 2014, Pimm et al. 2014, Johnson et al. 2017), with some parts of the world, and some taxonomic groups, such as marine fish assemblages, being particularly affected (Blowes et al. 2019). To understand how this biodiversity change is ocurring during the Anthropocene, we need to examine the mechanisms that underpin temporal change in communities (Rosenzweig 1995, McGill 2010, Blowes et al. 2019) and this at the spatial scale at which these

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drivers operate. One way ecologists can address this issue is by assuming that communities have a capacity, i.e. that they can support a certain number of individuals (or biomass). This capacity may be set by a series of factors that affect most of the species assemblage, including the amount of physical space available, habitat heterogeneity and primary productivity (Magurran et al. 2011). Following this reasoning, spatial diversity gradients may help to elucidate local biodiversity constraints when matched with the corresponding environmental gradients.

In recent decades, considerable progress has been made in describing marine fish diversity gradients and understanding their drivers. Much of this work has involved broad seascapes such as continental shelves (Ellingsen & Gray 2002, Anderson et al. 2013, Smoliński & Radtke 2017) as well as global-scale analyses (Tittensor et al. 2010, Edgar et al. 2017). Conversely, only a few studies have reported the existence of diversity gradients in mesoscale seascapes, such as bays or gulfs. Therefore, relatively little is known about the drivers behind such diversity gradients and how persistent they are over time (Rapoport 1994, Gaston & Blackburn 2008). Furthermore, given the lack of well-defined boundaries in the marine environment, gradients could be weak in communities composed of free-swimming species with high dispersal ability, such as fish assemblages (Olson et al. 1994, Soininen et al. 2007, Bergstad 2013).

As a direct consequence of environmental change during the Anthropocene era, demersal fishes are among the assemblages showing the highest rates of reorganization (Ellingsen et al. 2015, Magurran et al. 2015). Demersal fishes live in association with the seafloor, but can undertake wide horizontal and vertical migrations (Menni 1983, Bergstad 2013). Among other variables, the spatial variation of demersal fish diversity in macroscale seascapes is associated with latitude, ocean depth, temperature, salinity, oxygen, chlorophyll and their proximity to marine fronts (Ellis et al. 2000, Alemany et al. 2009, Anderson et al. 2013). Many of these variables are commonly used as proxies of productivity, which in turn is considered an indicator of available energy for biota to convert into biomass (Mittelbach et al. 2001, Evans et al. 2005). Energy availability is thought to be important in driving diversity patterns for all biological communities (Rosenzweig 1995, Gaston 2000, Mittelbach et al. 2001, Evans et al. 2005). The spatial structure of demersal communities comprising seascapes at mesoscales seems to be primarily responsive to bottom depth and substrate (Magnussen 2002, Busalacchi et al. 2010, Tamdrari et al. 2015). It has also been observed that some taxa in bays and gulfs, including fishes, show a diversity gradient that decreases with distance from the open sea. This pattern is known as the 'bay effect' (Rapoport 1994).

The aim of this study was to examine the shape and persistence of spatial relationships between α-diversity components of demersal fish assemblages and environmental drivers at the mesoscale. The focus of this work was the San Matías Gulf (SMG, Argentina), which is connected with the Patagonian Shelf Large Marine Ecosystem. It has a considerable bathymetric range and numerous sources of potential productivity, among which the most conspicuous is a seasonal frontal system that divides the gulf into 2 water masses, constituting 2 different habitats for demersal species. The SMG thereby provides us with an excellent opportunity for assessing, for the first time, the influence of the previously proposed drivers on local diversity patterns at the mesoscale. To that end, we made use of data available from 3 comprehensive fishery surveys conducted in different years, but during the same season and with the same sampling design.

2. MATERIALS AND METHODS

2.1. Study area

The SMG is a relatively young seascape; its basin was flooded approximately 11 000 yr ago during the post-glacial sea-level rise (Isla 2013). It is located between 40° 47′ and 42° 13′ S on the Atlantic coasts of South America, covering a total area of approximately 18 000 km^2 (Gagliardini & Rivas 2004; Fig. 1). The gulf is a semi-closed basin that has a limited water exchange with the adjacent shelf. It has 2 large, deep depressions between 160 and 216 m depth, one located south and one located north of the parallel 41°40' S. Approximately 55% of its total area is >100 m deep (Pierce et al. 1969, Piola & Scasso 1988). The mouth of the gulf is oriented north–south and is approximately 100 km long, with a 70 m sill (Gagliardini et al. 2005). Towards the east, the continental shelf forms an open basin with depths ranging between 45 and 80 m (Mazio & Vara 1983).

Different sources of biological production (primary and secondary production) in the gulf include: the tidal frontal system (Carreto et al. 1974, Rivas & Pisoni 2010, Williams et al. 2010), a zone of wind-driven upwelling events located off the west and north coasts (Carreto et al. 1974, Tonini et al. 2013, Pisoni et al. 2015), the San Antonio Bay (Esteves et al.

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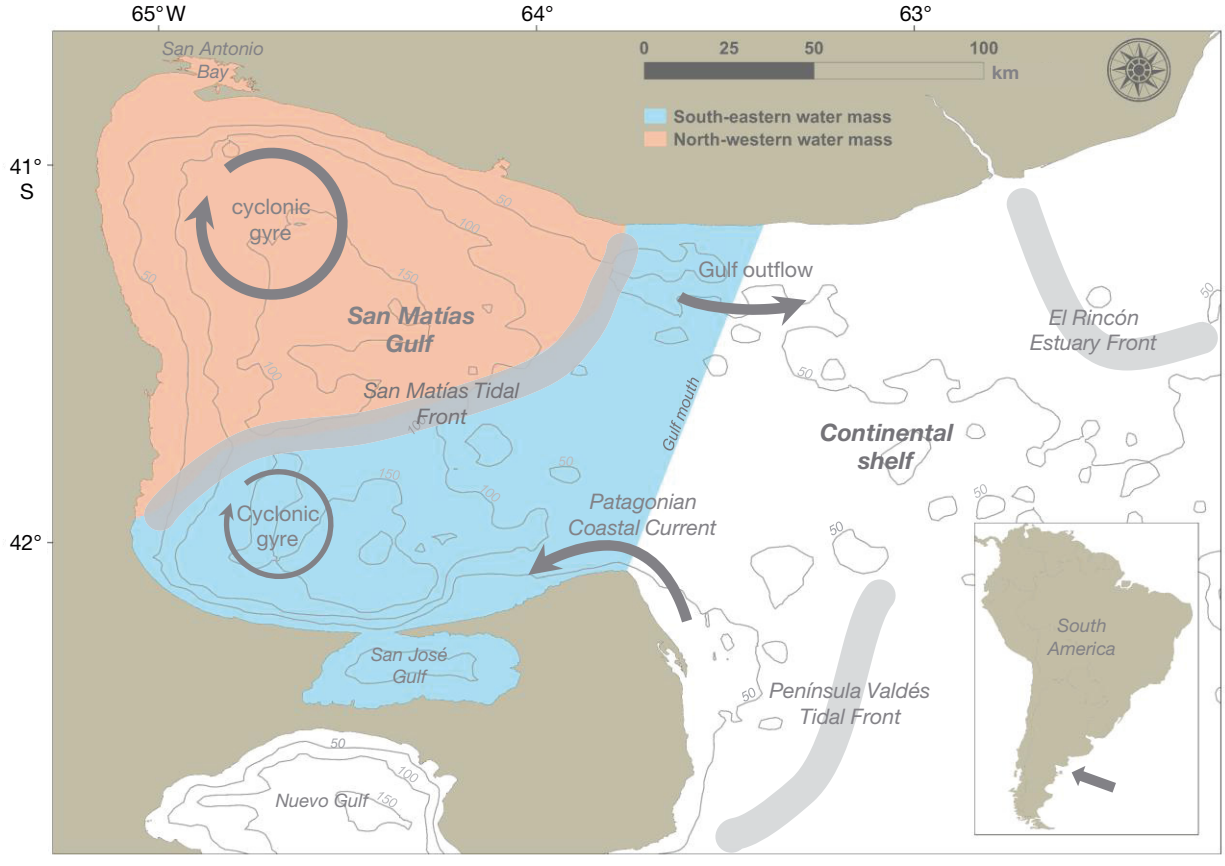

Fig. 1. North Patagonian Gulfs Region showing the study area (San Matías Gulf), a schematic drawing of the main frontal systems (gray thick lines) and the mean flow (gray arrows) (based on Piola & Scasso 1988, Acha et al. 2004, Tonini et al. 2013)

1996, Williams 2011), and the San José Gulf (Amoroso & Gagliardini 2010). Of these, the frontal system is the largest and is also the most studied oceanographic process in the SMG (Gagliardini et al. 2005, Ocampo Reinaldo et al. 2013, Romero et al. 2013). The SMG tidal front occurs from October until March (austral spring and summer) and divides the SMG into 2 water masses with different oceanographic conditions (Fig. 1). The northwestern mass has a well-defined thermocline, higher temperature and salinity, lower concentrations of nitrate and chlorophyll *a* (chl *a*) and a low water exchange rate. The southeastern mass has a well-mixed water column, lower values of temperature and salinity, higher concentrations of nitrate and chl *a*, and is strongly influenced by the inflow of cold water from the south (Carreto et al. 1974, Piola & Scasso 1988). When the tidal front is present, the sea surface temperature (SST) of the northwestern and southeastern water masses differs, on average, between 1 and 3°C. During the cold months (April to September), the front and the vertical stratification vanish, and the differences in SST are less than 1°C (Gagliardini & Rivas 2004, Williams et al. 2010).

2.2. Demersal species data and α-diversity indices

The demersal fish assemblage of the SMG consists of 40 species (22 Osteichthyes and 18 Chondrichthyes; Table S1 in the Supplement at www.int-res.com/articles/suppl/m638p149_supp.pdf) and is dominated by the Argentine hake *Merluccius hubbsi* (Romero et al. 2013). A preliminary study suggests that at least 6 compositional assemblages related with different depth ranges may by occurring in this gulf, but the Argentine hake constitutes more than 50% of the total abundance in all depths (Jeres et al. 2015).

Data on demersal fish species were collected during 3 fishery surveys in the spring (October–November) of 2006, 2007 and 2009 (Ocampo Reinaldo et al. 2011). The surveys, which were primarily designed to evaluate the abundance of the Argentine hake, followed a regular grid of 41 hauls and were conducted at depths between 30 and 160 m (Fig. S1). Using demersal trawling gear, the sampling scheme consisted of trawl hauls conducted with an average duration of 30 min at a speed of 3 knots and covered a mean swept area of 0.06 km^2 per haul. The mean latitudinal and longitudinal distance between hauls were 18.5 and 14.5 km,

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respectively. For each haul, a sample of 6 fish boxes (ca. 40 kg each) was taken at random from the total catch, and all individuals collected in the sample were identified to species level and counted.

Species richness and a PIE diversity metric (probability of interspecific encounter — a measure that is conceptually close to Simpson's diversity index) (Hurlbert 1971), which took species relative abundance into account, were used as response variables in order to evaluate the spatial structure of α-diversity in the SMG. Unlike richness, PIE is not sample-size dependent (Gotelli 2008). PIE was calculated for each haul (each sample of 6 boxes) as:

$$\mathrm{PIE} = \left(\frac{N}{N-1}\right)\left[1-\sum_{i=1}^{S}(p_i)^2\right] \quad (1)$$

where p_i represents the fraction of the total number of individuals that consists of species i, S is the total number of species in the haul and N is the total number of individuals in the haul. PIE is measured in units of probability, so it has values between 0 and 1 (Gotelli 2008).

Because the sample size in all hauls was the same (6 boxes), we checked whether the there was a statistical relationship between species richness with the total abundance of the haul through a linear regression analysis. It is important to note that the total abundance of the catch was obtained by simply multiplying the total number of boxes of the catch and the average number of individuals per box of the biodiversity sample.

2.3. Drivers

Eight variables were selected as potential drivers of the spatial structure of the demersal community in the SMG (Table 1, Fig. S1). These variables had been identified as main drivers of α-diversity of demersal assemblages in previous studies at different spatial scales (Table 1). In order to obtain a representative metric for the trawling area of each haul of all variables except depth, latitude and longitude, the following procedure was performed. First, the track lines of trawling events were reconstructed with the recorded locations at their beginnings and ends. These track lines were then rasterized to the same resolution of the environmental variables (i.e. MODIS Images resolution = 1000 m) and then the 'sample' tool in ArcMap 10.3 was used to extract the values of the environmental variables over each crossed pixel. Finally, a mean value of each variable for each haul was calculated using all of the pixel values corresponding to each haul line. Predictions, metrics, measurements and other methodological aspects for each driver are detailed in Table 1.

2.4. Analytical approach

We adopted a scale-explicit, multi-dimensional perspective on biodiversity and its variation (Chase et al. 2018). Different environmental variables were used as predictors, including an anthropogenic one (Table 1), and 2 components of biodiversity as response variables (richness and PIE) in a mesoscale seascape. In the marine realm, processes that influence diversity at the mesoscale extend over 10s to 100s of km and can last between hours and months (Mannocci et al. 2017). Mesoscale models allow the user to explore and answer ecological questions corresponding to environmental covariates, operating from fine- to macro-spatial scales, describing anything from short-term ocean variability to persistent patterns, and describing long-term ocean climates (Mannocci et al. 2017).

Fig. 2 summarises the main analytical steps conducted in the present work. We first interpolated the diversity indices with the inverse distance weight function as implemented in ArcMap 10.3 (https://desktop.arcgis.com/es/arcmap/). The number of neighbours used to compute the metric was set between 3 and 8. The resulting maps show the actual spatial patterns of diversity indices for each year and aid the interpretation of the final modelled patterns (Output 1 in Fig. 2). The second analytical step we undertook involved the calculation of Pearson correlation coefficients among all variables to ensure that they were not highly correlated, thus avoiding collinearity issues in the models ($r_{pearson} > 0.75$). This analysis was also used in the final interpretation of modelled patterns (Output 2 in Fig. 2).

The third analytical step was to model the relationships between diversity indices (richness and PIE) and predictors with boosted regression trees (BRTs; Output 3 in Fig. 2; Leathwick et al. 2006, Elith et al. 2008, Smoliński & Radtke 2017). This method proved particularly advantageous in the present study since it can be used to uncover dominant patterns between different potential drivers in a hierarchical way. Moreover, it can handle non-linear relationships and complex interactions, both of which are prevalent in ecological data (Guisan & Zimmermann 2000, Elith et al. 2008). BRTs constitute a powerful tree-based ensemble technique, combining the algorithms of regression trees with a machine-learning approach (i.e.

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Table 1. Variables, metrics and indicators used for each driver to describe predictions of demersal diversity in the San Matias Gulf (SMG) (Patagonia, Argentina)

Described environmental drivers and predictions	Metric, abbreviation, mean (range), [units]	Measurement
Contemporaneous primary productivity Primary productivity is considered by ecologists to be a proxy of available environmental energy. The most common patterns observed in aquatic communities are monotonic increase and unimodal (hump-shaped) relationships of diversity with available energy (Gaston 2000, Mittelbach et al. 2001, Chase & Leibold 2002). We believe that the prey of demersal fish is concentrated within the vicinity of areas of high primary productivity. For the SMG, it is therefore expected that demersal diversity peaks at medium to high levels of contemporaneous productivity (weekly resolution).	Mean daily water column primary production, P_{ZT}, 2175.4 (813.7–3213.8) [mg Cm^{-2} d^{-1}]	Primary productivity was calculated using a model that showed a reasonable performance in the Argentine sea (Platt & Sathyendranath 1988, Dogliotti et al. 2014). The model was run in SeaDAS 7.3 software and the input variables were chl *a*, photosynthetically available radiation (PAR) and the diffuse attenuation coefficient for downwelling irradiance at 490 nm. These variables were derived from MODIS Aqua Level-2 files corresponding to the dates of the surveys. These products were acquired from the NASA ocean color website (http://oceancolor.gsfc.nasa.gov). For each study year, the 3 best satellite images were selected and averaged over the periods in which the utilised fishery surveys were conducted (±15 d from the mean date of each survey).
Contemporaneous temperature SST is the environmental variable that best describes the oceanographic processes of the SMG and has been related to the distributions of the main demersal fish species (González et al. 2010, Williams et al. 2010, Ocampo Reinaldo et al. 2013). Therefore, in the present study, SST was considered an indicator of the 2 different water masses, which in turn might indicate different habitats for demersal fishes.	Mean sea surface temperature, SST, 12.81 (6.22–14.82) [°C]	For each year, the 3 best SST satellite images (MODIS-NASA) were selected and averaged over the periods in which fishery surveys were conducted (±15 d from the mean date of each survey).
Proximity to marine fronts Marine fronts are areas of high biological production and are characterised by food availability at depth and high larval retention for demersal fishes (Acha et al. 2015). Demersal diversity has been observed to be higher at seasonal fronts than in the adjacent, non-frontal zones (Alemany et al. 2009). In the present work, we predicted that the relationship between demersal diversity and proximity to fronts would resemble one of the 2 most common forms of the species–energy relationships (i.e. monotonic increase or hump-shaped).	Distance to the main surface thermal front, Distance to front, 32019 (0–93932) [m]	For each year, an edge layer from the single best SST image was produced using the Cayula-Cornillon algorithm, as implemented in MGET package (Cayula & Cornillon 1992, Roberts et al. 2010). The main settings of this algorithm were: front detection threshold = 0.1°C km^{-1}, histogram window size = 64 pixels and histogram window stride = 10. The other parameters were used with default values. The resulting edge rasters were transformed to polygons. The distance from this feature was calculated using the Spatial Analyst extension in ArcMap 10.3.
Depth At depths lower than 200 m (the bathymetry range of SMG), peaks of α-diversity were reported between 0 and 50 m (Smoliński & Radtke 2017), between 20 and 40 m (Moore et al. 2010) and between 80 and 120 m (Gaertner et al. 2005). These studies attributed these patterns to other factors that vary with depth, such as dissolved oxygen, food availability and substrate. Thus, different patterns were expected to emerge between demersal diversity and depth in the SMG. However, considering that the photic zone is a source of food and oxygen, and that distance to the euphotic zone increases with depth, we predicted that negative relationships between diversity and depth were more likely to occur.	Mean depth of the haul, depth, 128.6 (32.1–170.7), [m]	The depths registered by echo sounders at the start and end of each haul were averaged.

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Table 1 (continued)

Described environmental drivers and predictions	Metric, abbreviation, mean (range), [units]	Measurement
Trawling fishery disturbance		
The trawling fleet of the SMG is primarily directed to the Argentine hake (80% of annual landings) year-round and secondarily to the silver warehou *Seriolella porosa* during August and September. The trawling fleet operates between 40 and 200 m depth, with the highest fish intensity occurring at around 160 m. Fishers follow a process of 'trial and error' and the success of a fishing haul generates repetitions within the same area. Conversely, a haul with a small catch of the target species motivates the search of a different area (Ocampo Reinaldo et al. 2013). Fishing affects demersal fish communities in many ways (Bianchi et al. 2000) and is considered one of the most important drivers of biodiversity in marine ecosystems (Bianchi et al. 2000, Worm et al. 2006, Ellingsen et al. 2015). We expected differences in α-diversity between places presenting different intensities of trawling activity.	Density of trawl fishing hauls, haul density, 6.43 (0–26.954), [no. hauls km^{-2}]	The variable 'density of trawl fishery hauls' was calculated for a regular grid of 1000 m of cell resolution, covering the entire study area over the period 2004–2008 of data availability. Data on trawl fleet activity were gathered by a vessel monitoring system (González et al. 2004). Methodological procedures for variable construction are detailed in Ocampo Reinaldo et al. (2013).
Local connectivity with the open ocean (bay effect)		
In gulfs, the expectation is that α-diversity decreases with an increase in distance to the open sea (Rapoport 1994). We expected diversity to increase proportionally with proximity to the adjacent continental shelf.	Longitude, long, (64.85°–64°W)	The longitude of the middle position of each haul was used. The mouth of the SMG is oriented in a north–south direction, and the coefficient of correlation between distance to mouth (i.e. distance to the continental shelf) and longitude is almost 1 (G. M. Svendsen unpubl.).
Latitude		
At large latitudinal extensions (more than 15 latitudinal degrees), different forms of the relationship between α-diversity of demersal communities and latitude have been described. These relationships are mediated by other factors that correlate with latitude, such as sea temperature (Lucifora et al. 2012, Anderson et al. 2013). The SMG covers a latitudinal range of less than 2°, therefore the influence of this variable on demersal diversity is not expected to be strong.	Latitude, lat, (42.18°–41.0° S)	The latitude of the middle position of each haul was used.
Survey year		
Interannual changes in biodiversity are expected to occur due to drivers which were not tested in the present study.	Study year, year; 2006, 2007, 2008	Years in which surveys were conducted.

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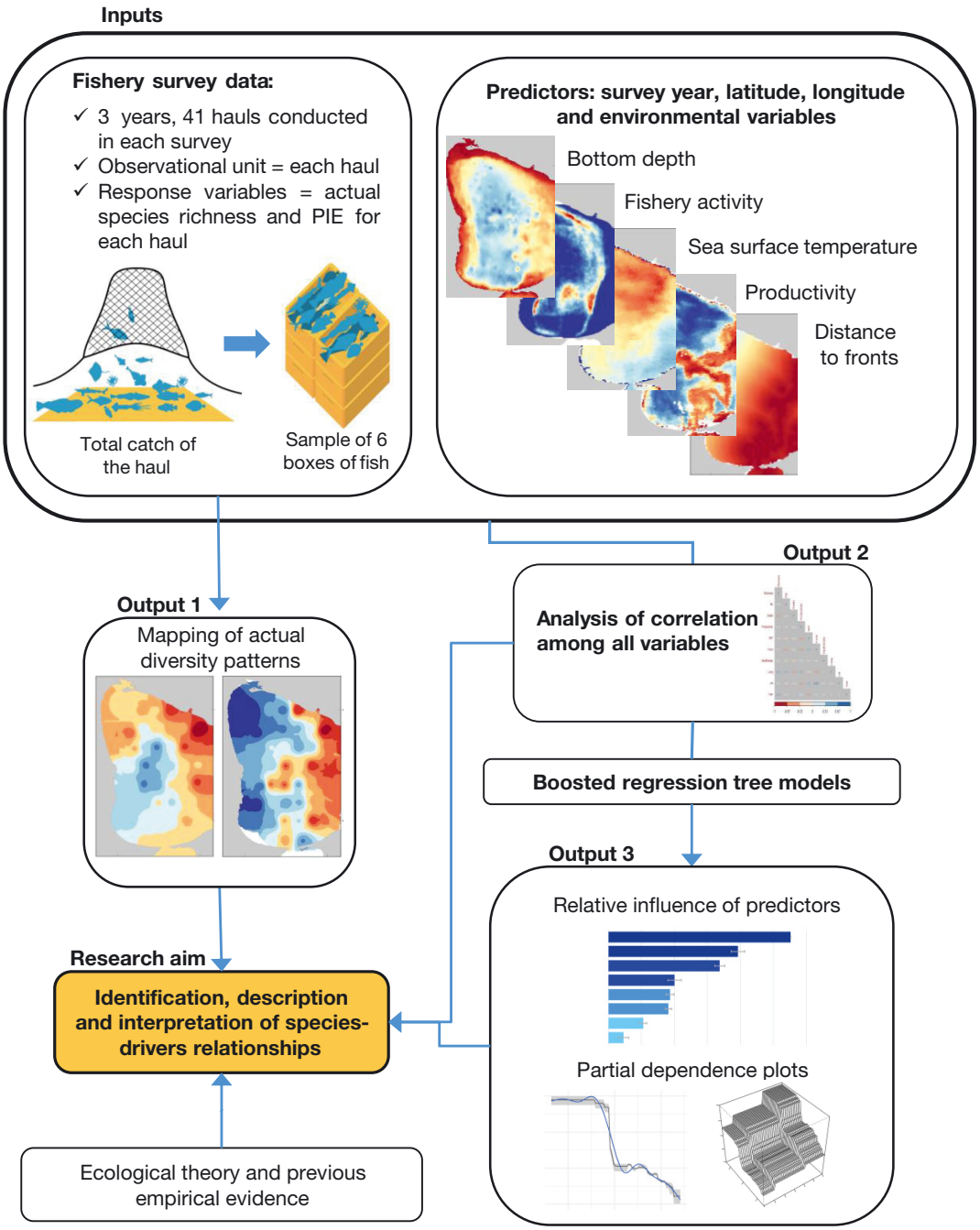

Fig. 2. Flow diagram of the analytical approach used in the present study

boosting) in order to improve model accuracy. Through boosting, simple tree models can be fitted iteratively using a forward stepwise procedure, minimising the loss of predictive deviance at each step (Leathwick et al. 2008, Elith et al. 2008). Each successive tree fits the residuals from the previous model, and the final model is a linear combination of all the trees (Elith et al. 2008). The BRT technique automatically incorporates interactions between predictors; it can effectively select relevant variables, can handle categorical as well as continuous input variables and allows collinearity between environmental variables (Elith et al. 2008). When modelling non-linear effects between response and predictor variables, BRTs have proven to perform as well as, or better than, other traditional models, such as generalized linear or generalized additive mixed models (GLMMs/GAMMs), and they are not subject to the caveats of these techniques (Elith et al. 2006, De'ath 2007, Martínez-Rincón et al. 2012).

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BRT models require the user to set 'tuning' parameters to avoid overfitting the model. The parameters that most influence variable selection and model performance are learning rate, bag fraction and tree complexity. Learning rate (shrinkage parameter) limits the contribution of each tree to the overall model explanation, while the tree complexity (number of splits of each tree) represents the order of the interactions between the predictors that would be modelled. Learning rate and tree complexity influence the number of trees required to reach the optimal model performance. Bag fraction is the proportion of data selected to build the model at each step (e.g. a bag fraction of 0.5 means that 50% of the data are randomly selected without replacement to fit a tree; Elith et al. 2008). This parameter introduces stochasticity in BRT model building, reducing overfitting and improving accuracy (Friedman 2002). The total number of trees in the model is then optimized using cross-validation (CV). CV proceeds by randomly selecting a proportion of the data to fit the model (training set) and then testing predictive performance using the data left out (out-of-bag data) of the fitting process (testing set).

Optimal BRT models were selected using the following parameterization setting: learning rate = 0.001, in order to allow each model to reach at least 1000 trees following the 'rule of thumb' for balancing model complexity and performance (Elith et al. 2008, Soykan et al. 2014); tree complexity = 3, given the relatively low sample size; and bag fraction = 0.75. These settings avoid model overfitting and attain the highest accuracy, as indicated by the lowest values of CV deviance and standard error (Elith et al. 2008). All BRTs were modelled with the default 10-fold CV to determine the optimal number of trees for each model. The BRT for richness was fitted based on a Poisson distribution while the PIE model was fitted based on Gaussian distribution. Performance was described by the cross-validated proportion of the total deviance explained (D^2), which provides a measure of the goodness-of-fit between the predicted and raw values (Leathwick et al. 2008).

Model interpretation was carried out by the estimation of the relative influence of predictors on the response (Output 3 in Fig. 2), with higher numbers indicating stronger influence on the modelled response (contributions were scaled to sum to 100). The variable influence is based on the number of times each variable was selected for splitting, weighted by the squared improvement at each split, and then averaged over all trees. In order to account for uncertainty around the estimates, the explained deviance and the relative influence of each explanatory variable was calculated on 500 bootstrap simulations of each BRT model. Similarly, the accuracy of our predictive model was evaluated calculating 95% confidence intervals of model predictions on the response variable. The final BRT fitted functions were visualized with partial dependence plots with smoothing, which show the effect of predictor variables on the response variable (output 3 in Fig. 2; Elith et al. 2008). Finally, the strength of all 2-way interactions in models was examined using the GBM interactions function. Models were built in R software, v. 3.5.1 (R Core Development Team 2018) with the 'gbm' (Ridgeway 2006) and 'dismo' (Hijmans et al. 2017) packages and customized code such as in Elith et al. (2008).

3. RESULTS

3.1. Spatial patterns of actual α-diversity

The estimated fish abundance per haul ranged between 117 and 23 154 individuals (mean = 2318 individuals). We did not find a significant trend of the species richness in relation to haul total abundance (df = 122, F = 0.006299847, p = 0.93; Fig. S2). A total of 40 demersal fish species (Table S1) were recorded across the 3 surveys and all the hauls. Actual richness per haul ranged from 6 to 18 in 2006, from 6 to 21 in 2007, and from 9 to 25 in 2009 (Fig. 3). PIE per haul ranged from 0.14 to 0.80 in 2006, from 0.07 to 0.78 in 2007, and from 0.05 to 0.82 in 2009 (Fig. 3). While the observed spatial patterns of both indices varied over the years, a general gradient from west to east emerged each year (i.e. hauls with highest diversity were mainly located close to the gulf mouth, while hauls with lowest diversity were located mainly in the innermost part of the gulf; Fig. 3).

3.2. BRT models

As no strong collinearity was detected between any pairs of variables (all $r_{pearson}$ where lower than 0.75; Fig. S3), all variables were included in the analyses. The final models accounted for the 8 predictor variables which had non-zero influence, BRT models for both response variables had a good predictive power (D^2 > 30%; Table 2), and the relationships between observed and predicted diversity metrics were relatively high (both regression analyses were statistically significant at p < 0.01, and regression coefficients were $R^2_{richness}$ = 0.79, R^2_{PIE} = 0.73;

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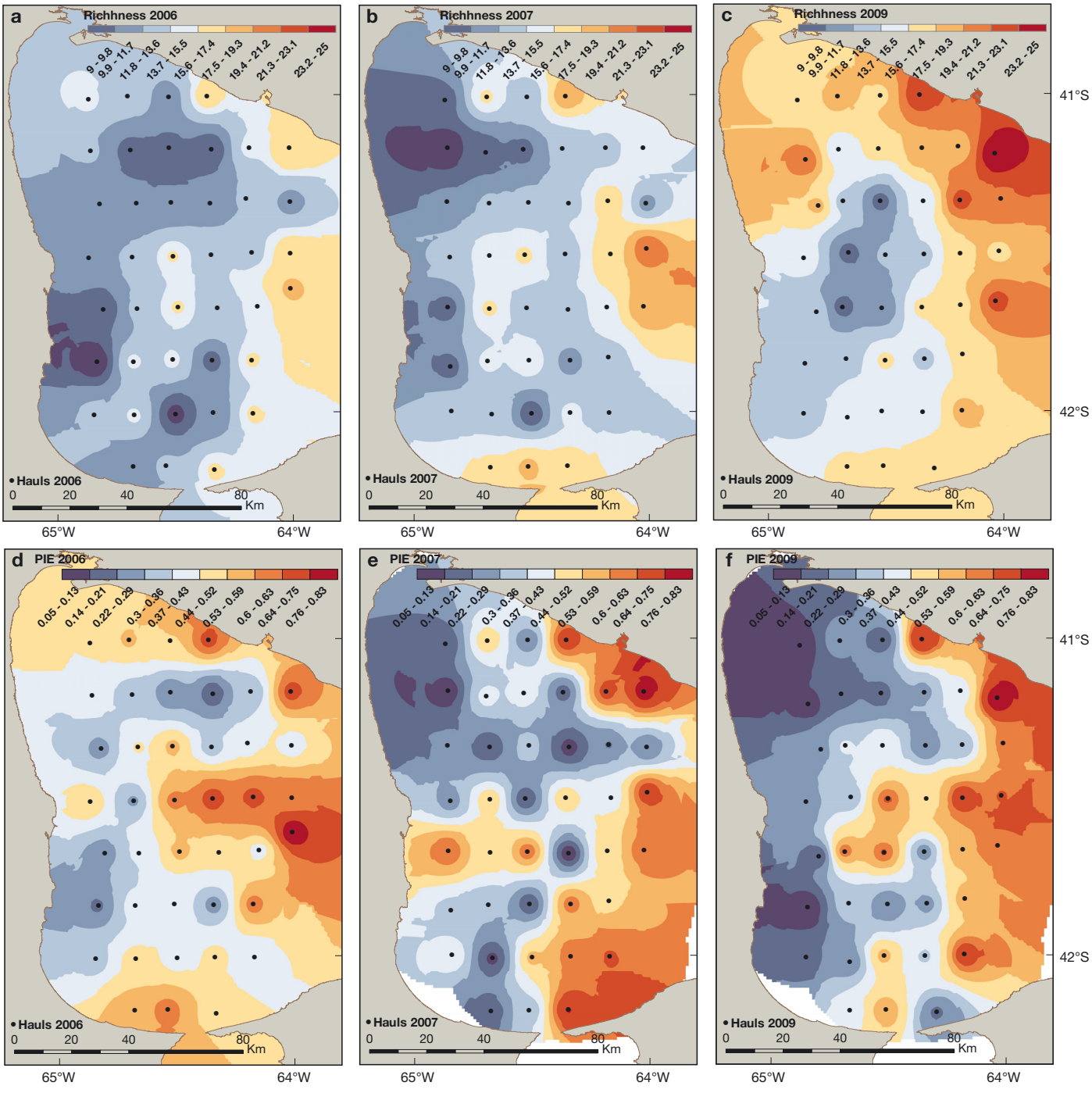

Fig. 3. Inverse distance weighting interpolation maps of (a–c) actual richness and (d–f) probability of an interspecific encounter (PIE) in 2006, 2007 and 2009. Colour palette corresponds to 8 quantiles for each variable in each year. Black points correspond to centroids of haul tracks in each survey

Fig. S4). These results indicate that both BRT models provided a good fit for the observed data set.

Fig. 4 shows the resulting influence hierarchies of predictors on BRT models for each diversity response and Figs. 5 & S5 show the bidimensional partial dependence plots of diversity indices, for all predictors, arranged in the same hierarchies. Depth was the most important variable for explaining variation in both response variables. The highest predicted values for species richness occurred between 40 and 80 m depth, then decreased abruptly to 100 m and thereafter slightly decreased towards 170 m. The highest

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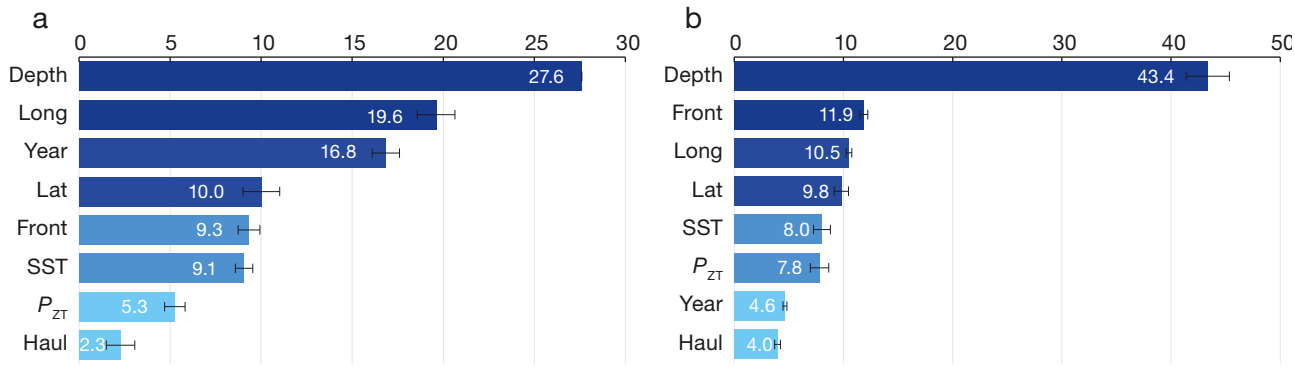

Fig. 4. Mean ± SD relative influence of drivers (%) as ranked by the final boosted regression tree (BRT) model (resulting from 500 parametric bootstrap simulations of each BRT model). Explanation of the variables and units are provided in Table 1. (a) Species richness model. (b) PIE diversity model

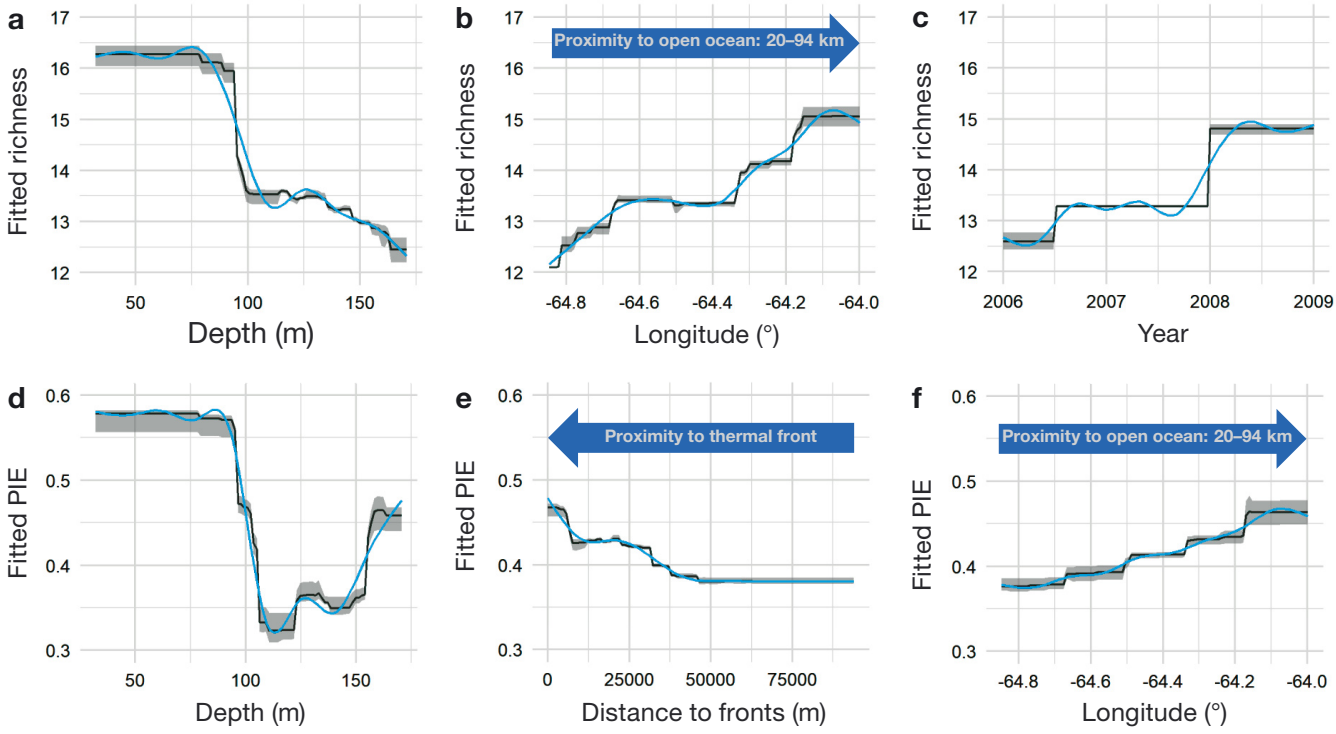

Fig. 5. Partial dependence plots showing the predicted response for (a–c) richness and (d–f) PIE versus the 3 main predictor variables. Black line: fitted function; grey ribbon: confidence intervals calculated by 500 parametric bootstrap simulations of each BRT model; blue line: smoothed trend. Values were predicted for each variable showed in the axes, keeping the values for all other predictors at their mean value. For details about predictor variables, refer to Table 1

values for PIE occurred between 40 and 100 m depth, thereafter abruptly decreased at 120 m, and from there began to increase stepwise towards 160 m, the maximum depth. Longitude was the second-most important predictor of richness and the third-most important predictor of PIE. Both diversity indices increased almost linearly towards the east, i.e. towards the mouth of the SMG and the open ocean. Distance to front was the second-most important variable for explaining variation in PIE. The maximum values coincided with the front (distance = 0 km) and then linearly decreased, reaching the minimum approx. 50 km away from it, then remained constant until the maximum distance. Distance to front had a medium influence on variation in richness. Richness reached its lowest value at the front, from where it rose at approx. 25 km, then decreased, until it reached a new minimum at 50 km, thereafter increasing to a maximum between 60 and 90 km. Year was the third variable in order of influence on species richness, which increased between the first and the last study year. PIE showed the opposite pattern, but the influence of

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year on this index was quite low. Latitude had a medium influence on both diversity indices. Species richness exhibited an irregular pattern in relation to latitude whilst PIE showed a unimodal pattern, with the hump located at mid latitudes of the gulf. A similar but muted variation of both diversity indices occurred in relation to temperature, productivity and haul density. Both indices decreased slightly at the highest temperatures (between 12 and 15°C), increased slightly at the highest productivity values (P_{ZT} > 2500 mg C m^{-2} d^{-1}) and increased slightly at low levels of haul density (<10 hauls km^{-2}) (Fig. 5).

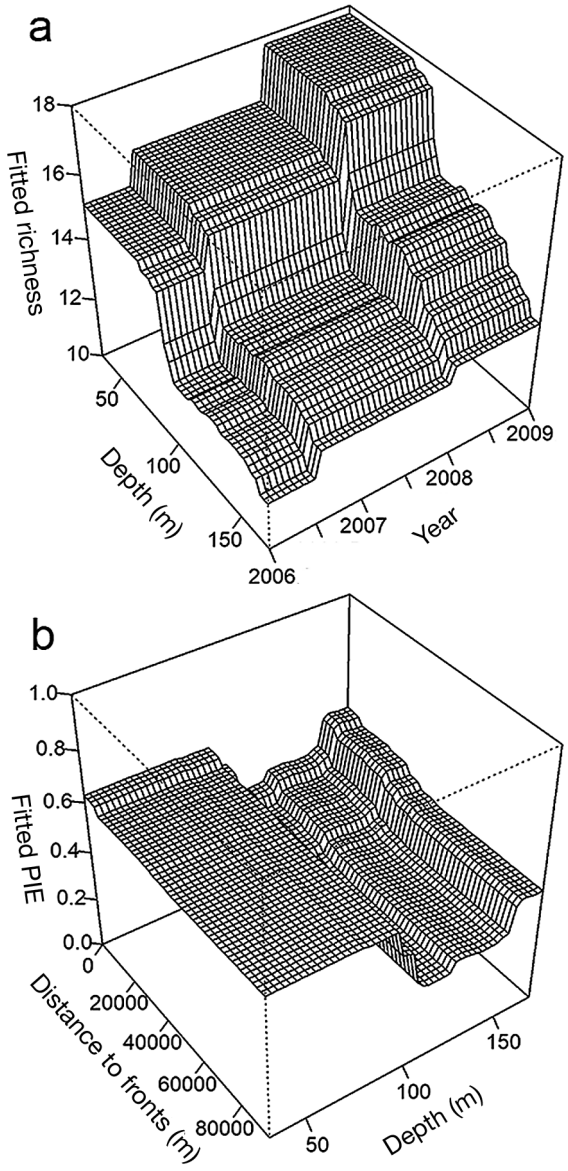

Fig. 6. Three-dimensional partial dependence plots for (a) species richness and (b) PIE. Values were predicted for each variable shown in the axes, keeping the values for all other predictors at their mean value

In all models, interaction between predictors played an important role. BRTs that modelled interactions between predictors performed much better than simpler BRT models that excluded interactions (i.e. tree complexity = 1; CV deviance explained in richness model = 22%, CV deviance explained in PIE model = 27%). The most important pairwise interactions in the richness model occurred between depth and year, and between latitude and year (Table S2); in 2009, a differential increase of richness was observably greater at depths less than 100 m (Fig. 6a). The increase of richness in 2009 was greater when north of −41.6° latitude. The most important interaction in the PIE model occurred between distance to fronts and depth (Table S2, Fig. 6b). The increase in PIE with proximity to frontal zones was steepest at depths greater than 100 m (Fig. 6b). The remaining interactions were all of relatively minor importance, with interaction strengths one order of magnitude lower than the top-ranked interaction. Although both diversity indices changed over the years, the shape of the relationships with predictors was persistent during the period studied (Figs. 6a & S6).

4. DISCUSSION

To the best of our knowledge, this report on the spatial structure of local demersal assemblages in a mesoscale seascape, driven by static and seasonal dynamic variables, is the first of its kind to operate at so high a resolution. Over a study period of 4 yr, we found persistent diversity gradients of species richness and PIE. These patterns seemed to be driven by 3 main environmental gradients: bottom depth, local connectivity with the open ocean, and proximity to the thermal front (Figs. 5 & 6). The modelled patterns were less significantly affected by the other drivers included in the analysis. To our knowledge, the persistence through years of spatial patterns of α-diversity have never been considered in studies of marine fish assemblages. It has been noted that some mesoscale seascapes in the Mediterranean Sea which are hotspots of demersal fish diversity could persist in the same location for at least a decade to come (Garofalo et al. 2007, Busalacchi et al. 2010). Very different geographical distribution arrangements have been observed in the Gulf of Saint Lawrence over a 20 yr period with regard to the relocation of hotspots for demersal fish diversity (Tamdrari et al. 2015). These observations suggest that stability of the spatial structure of α-diversity in demersal fish assemblages might vary considerably with

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each different seascape. The stability of environmental gradients, which in turn drive diversity gradients, should be considered when assessing the impact of anthropogenic factors or environmental change in demersal fish communities. A recent study showed that community-level regulation involves both species richness and total abundance (Gotelli et al. 2017). The regulation of species diversity implies the maintenance of shared, key resources and connectivity with an open system (Brown et al. 2001, Gotelli et al. 2017). In order to better understand community regulation, it is therefore necessary to study the stability of the diversity gradients derived from these processes. To this end, our results constitute a starting point for showing that the shape of diversity gradients can persist irrespective of interannual changes in species number on a regional scale (concerning the entirety of the SMG).

4.1. Depth patterns

Bottom depth is one of the main factors explaining spatial variation in diversity of demersal assemblages. However, different patterns emerged in different seascapes and thus no general rule exists (Rosenzweig 1995, Magnussen 2002, Gaertner et al. 2005, Busalacchi et al. 2010, Moore et al. 2010, Anderson et al. 2013, Tamdrari et al. 2014, Acha et al. 2015). Researchers associate depth patterns with several variables that covary with the bathymetric gradient, and which directly constrain fish distribution. These include: temperature, oxygen concentration and food supply (Rosenzweig 1995, Gaertner et al. 2005, Tamdrari et al. 2015). Recent studies identify dissolved oxygen as the better depth covariate for constraining the distribution of demersal assemblages (Lucifora et al. 2012, Sperling et al. 2016); they found that diversity decreased with oxygen concentration. In light of these results, we conducted an ad hoc exploration of historic measurements of dissolved oxygen in the SMG and found that this variable remains high and almost constant at 40 to 50 m depth and then decreases linearly towards 160 m (Fig. S7). This trend is similar to the patterns we found between decreasing richness and PIE with depth (Fig. 5a,d). Interestingly, the oxygen concentration at 160 m in the SMG corresponds to the median sublethal threshold described for benthic fishes (i.e. 4.4 mg l^{-1}; Fig. S7; Vaquer-Sunyer & Duarte 2008). This suggests that oxygen should be a limiting resource for some species in the deepest areas of the SMG. A greater abundance of a few fish species, better adapted to bottom conditions, and a decrease in abundance of those that thrive elsewhere, could therefore explain the increase in PIE in the deepest areas (>120 m) of the SMG.

4.2. Connectivity with the open ocean (bay effect)

The demersal richness of the SMG decreases with distance from the adjacent continental shelf, a pattern already proposed for some gulfs and bays (Rapoport 1994). This pattern is called the 'bay effect' and seems to be a consequence of isolation — a reduction in colonization rates with distance from the source pool of immigrants (i.e. the largest connected area, the open ocean for a gulf; Rapoport 1994, MacArthur & Wilson 1967, Gaston & Blackburn 2008). We found that PIE also reflects the 'bay effect' pattern and, to our knowledge, this is the first time that it has been reported in a marine fish assemblage. The presence of these patterns may indicate that the SMG is prone to colonization by new species. This agrees with a recent study which showed that the SMG is an ecosystem that is still maturing and, as a consequence, has niches yet to be occupied (Ocampo Reinaldo et al. 2016).

The 'bay effect' of the SMG may be being reinforced by 2 frontal systems located just outside of its mouth: the El Rincón estuary front and the Península Valdés tidal front (Acha et al. 2004). These fronts are considerably larger than the SMG tidal front and may be acting not only as species sources, but also as productivity sources. These 2 systems delimit an ecotone between 2 biogeographical provinces of the Argentine sea (Balech & Ehrlich 2008), just outside of the SMG mouth, within which is located one of the main hotspots of Chondrichthyes of the southwest Atlantic (Lucifora et al. 2012). This hotspot may be supplying species to the SMG. In addition, the SMG receives water, rich in chlorophyll and nutrients from the Península Valdés front via the southern sector of its mouth (Amoroso & Gagliardini 2010, Williams et al. 2010), generating the high values of productivity that are observed in the east sector of the gulf (see productivity maps in Fig. S1). We detected a slight increase in both indices of diversity at high values of contemporaneous productivity. The hauls conducted over the productive waters of the southeast sector of the gulf (Fig. 3) are most likely responsible for these patterns.

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4.3. Proximity to front patterns

We discovered a relationship between the monotonic increase of PIE and the increasing proximity to the marine front. A weaker unimodal pattern was found between species richness and distance to the front. Fronts are typically associated with high primary and secondary production and are considered sources of energy; offering better feeding opportunities, spawning grounds and migration routes for demersal fish (Olson et al. 1994, Bakun 1996, Acha et al. 2015). Diversity was therefore expected to increase linearly with proximity to marine fronts, or first increase and then decrease (i.e. a unimodal or 'hump-shaped' relationship; Gaston 2000, Mittelbach et al. 2001, Chase & Leibold 2002). However, the effect of fronts on demersal diversity is not clear yet, and this could be because demersal fishes are highly mobile species and the few studies that addressed this relationship were conducted at broad spatial scales. Leathwick et al. (2006) used a grid of 4 km resolution and observed that species richness was higher in areas with high gradients of SST and areas with high primary productivity and attributed these patterns to the subtropical front of New Zealand. Alemany et al. (2009) found that richness and PIE were higher in large zones (>20 000 km^2) in which a seasonal front occurs than in large zones without fronts. In the same study, they also found a negative relationship between demersal diversity and permanent fronts. In contrast, Lucifora et al. (2012) used a grid of 1° × 1° and found that hotspots of chondrichthyan diversity in the southwest Atlantic were associated with seasonal and permanent marine fronts.

Here, the survey design we employed enabled us to evaluate the effect of the front, at a lag of 16.5 km (i.e. the mean distance between consecutive hauls), over a smaller spatial extension and at a higher resolution (1 km) than generally examined. We found that the seasonal front influenced both PIE and richness at distances of up to 50 km. Fronts can therefore be considered sources of shared resources that drive spatial structure in demersal assemblages. Our results also revealed that the effect of the front on PIE is strongest in deeper areas. This pattern suggests that the few species adapted to bottom conditions benefit from resources supplied to them by fronts, and they are more abundant as a result. Fronts are responsible for vertical fluxes that transport food particles to the seafloor, thereby enhancing food availability for benthic and demersal species (Acha et al. 2015). Therefore, the food supplied by the SMG front appears to allow demersal fishes to be present in deep places far from the euphotic zone where, consequently, no other food sources are available.

4.4. Latitudinal patterns

Latitudinal patterns of demersal diversity were observed in previous studies at very large latitudinal ranges (Lucifora et al. 2012, Anderson et al. 2013). Due to its short latitudinal range, we did not expect any strong effect of this variable within the SMG. Latitude nevertheless proved to be a moderate influence in both response variables: each peaked at different latitudes (Fig. S5). Since this relationship was not linear, it probably resembles the effects of local or zonal drivers. Latitude interacted with almost all other predictors in both BRT models and correlated, to a considerable extent, with distance to fronts (Fig. S3), hindering the interpretation of spatial patterns. In spite of these complications, the inclusion of latitude in the BRT models proved to be beneficial, since it allowed us to obtain a more consistent range of fitted diversity values in relation to the observed values.

4.5. Temperature patterns

Both indices of diversity showed a slight decrease with increasing SST. This pattern suggests a lower diversity over the northwest water mass, which is warmer, saltier and stratified, which may diminish vertical transport of food and resources to the bottom. This pattern was consistent with the one from a larger-scale study conducted in New Zealand by Leathwick et al. (2016). They found that fish richness is generally lower in warmer and/or more-saline waters than in cooler less-saline waters, and the latter being produced in zones of water mixing.

4.6. Fishery disturbance

Our results suggest that trawling activity slightly reduces richness and PIE, but these patterns may also reflect the fact that most fishery hauls were conducted in places where the dominance of the Argentine hake *Merluccius hubbsi* was very high. Numerous studies suggest that Argentine hake reduces fish diversity richness in areas where it is very abundant (Angelescu & Prenski 1987, Alemany et al. 2009, Ocampo Reinaldo et al. 2011). Nonetheless, in order to address with accuracy the effect of trawling on demersal diversity, a longer study period is needed.

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4.7. Final remarks

Our results indicate that around one-third of the variation of each diversity index could be explained by a combination of strong environmental drivers (Table 2). The remaining two-thirds may be the result of stochasticity (McGill 2010), sampling issues inherent to the fishing method and other drivers not addressed in the present work. BRT models did not successfully predict the most extreme values of both response variables (Fig. S4). We believe that such values are influenced by local drivers not considered in our models. The hauls conducted in the northeast extreme, for example (first northern latitudinal line and last eastern longitudinal column of the survey design; Fig. 3), showed the highest values of diversity in all years. They were located in close proximity to a conspicuously rocky reef, known by fishermen as a net-hitch zone, which provides refuge and feeding opportunities for many fish.

The maintenance of environmental gradients, specifically those related to shared resources and connectivity to an open system, appears to be essential for safeguarding the stability of demersal fish communities. The study of these gradients may help to elucidate the processes behind diversity change trends across biogeographic regions. For instance, tidal fronts, which are common in temperate seas (Acha et al. 2015), may be linked to richness change and high rates of turnover in marine fish assemblages (Blowes et al. 2019). Such tidal fronts could also influence the resilience of temperate marine fish assemblages. Finally, our approach is relevant to other seascapes, since the key factors identified by this study are also present at smaller and larger scales than the mesoscale.

Table 2. Main parameters and performance of the boosted regression tree (BRT) models for richness and probability of an interspecific encounter (PIE). Estimates of predictive performance were calculated using 10-fold cross validation

BRT model	Richness	PIE
Error distribution	Poisson	Gaussian
Number of trees	6300	4100
Predicted response range	9.53–22.17	0.1790–0.6968
Total deviance	0.944	0.04
Predicted deviance, mean ± SD	0.604 ± 0.061	0.027 ± 0.0036
Deviance explained, mean ± SD (D^2; %)	31.8 ± 2.4	30.6 ± 2.5

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LITERATURE CITED

Acha EM, Mianzan HW, Guerrero RA, Favero M, Bava J (2004) Marine fronts at the continental shelves of austral South America: physical and ecological processes. J Mar Syst 44:83–105

Acha EM, Piola A, Iribarne O, Mianzan H (2015) Ecological processes at marine fronts: oases in the ocean. Springer, Cham

Alemany D, Acha EM, Iribarne O (2009) The relationship between marine fronts and fish diversity in the Patagonian Shelf Large Marine Ecosystem. J Biogeogr 36: 2111–2124

Amoroso RO, Gagliardini DA (2010) Inferring complex hydrographic processes using remote-sensed images: turbulent fluxes in the Patagonian gulfs and implications for scallop metapopulation dynamics. J Coast Res 26:320–332

Anderson MJ, Tolimieri N, Millar RB (2013) Beta diversity of demersal fish assemblages in the North-Eastern Pacific: interactions of latitude and depth. PLOS ONE 8:e57918

Angelescu V, Prenski LB (1987) Ecología trófica de la merluza común del Mar Argentino (Merlucciidae, *Merluccius hubbsi*). Dinámica de la alimentación analizada sobre la base de las condiciones ambientales, la estructura y las evaluaciones de los efectivos en su área de distribución. Serie Contribuciones del Instituto Nacional de Investigación y Desarrollo Pesquero (INIDEP), no. 561

Bakun A (1996) Patterns in the ocean: ocean processes and marine population dynamics. California Sea Grant College System/NOAA/Centro de Investigaciones Biológicas del Noroeste, La Paz

Balech E, Ehrlich MD (2008) Esquema biogeográfico del mar Argentino. Rev Invest Dessaro Pesq 19:45–75

Bergstad OA (2013) North Atlantic demersal deep-water fish distribution and biology: present knowledge and challenges for the future. J Fish Biol 83:1489–1507

Bianchi G, Gislason H, Graham K, Hill L and others (2000) Impact of fishing on size composition and diversity of demersal fish communities. ICES J Mar Sci 57:558–571

Blowes SA, Supp SR, Antão LH, Bates A and others (2019) The geography of biodiversity change in marine and terrestrial assemblages. Science 366:339–345

Brown JH, Ernest SM, Parody JM, Haskell JP (2001) Regulation of diversity: maintenance of species richness in changing environments. Oecologia 126:321–332

Busalacchi B, Rinelli P, De Domenico F, Profeta A, Perdichizzi F, Bottari T (2010) Analysis of demersal fish assemblages off the Southern Tyrrhenian Sea (central Mediterranean). Hydrobiologia 654:111–124

Carreto JI, Verona CA, Casal AB, Laborde A (1974) Fitoplancton, pigmentos y condiciones ecológicas del Golfo

Author copy

San Matías I. Campaña SAO I (Marzo 1971) Instituto de Biología Marina de Mar del Plata, Contribución no. 235, p 1–22
Cayula JF, Cornillon P (1992) Edge detection algorithm for SST images. J Atmos Ocean Technol 9:67–80
Chase JM, Leibold MA (2002) Spatial scale dictates the productivity–biodiversity relationship. Nature 416:427–430
Chase JM, McGill BJ, McGlinn DJ, May F and others (2018) Embracing scale-dependence to achieve a deeper understanding of biodiversity and its change across communities. Ecol Lett 21:1737–1751
De'ath G (2007) Boosted trees for ecological modeling and prediction. Ecology 88:243–251
Dogliotti AI, Lutz VA, Segura V (2014) Estimation of primary production in the southern Argentine continental shelf and shelf-break regions using field and remote sensing data. Remote Sens Environ 140:497–508
Dornelas M, Gotelli NJ, McGill B, Shimadzu H, Moyes F, Sievers C, Magurran AE (2014) Assemblage time series reveal biodiversity change but not systematic loss. Science 344:296–299
Edgar GJ, Alexander TJ, Lefcheck JS, Bates AE and others (2017) Abundance and local-scale processes contribute to multi-phyla gradients in global marine diversity. Sci Adv 3:e1700419
Elith J, Graham CH, Anderson RP, Dudík M and others (2006) Novel methods improve prediction of species' distributions from occurrence data. Ecography 29:129–151
Elith J, Leathwick JR, Hastie T (2008) A working guide to boosted regression trees. J Anim Ecol 77:802–813
Ellingsen K, Gray JS (2002) Spatial patterns of benthic diversity: is there a latitudinal gradient along the Norwegian continental shelf? J Anim Ecol 71:373–389
Ellingsen KE, Anderson MJ, Shackell NL, Tveraa T, Yoccoz NG, Frank KT (2015) The role of a dominant predator in shaping biodiversity over space and time in a marine ecosystem. J Anim Ecol 84:1242–1252
Ellis JR, Rogers SI, Freeman SM (2000) Demersal assemblages in the Irish Sea, St George's Channel and Bristol Channel. Estuar Coast Shelf Sci 51:299–315
Esteves JL, Solís M, Sastre V, Santinelli N, Gil M, Commendatore M, González Raies C (1996) Evaluación de la contaminación urbana de la Bahía San Antonio. Fundación Patagonia Natural, Chubut (Argentina) Plan de Manejo Integrado de la Zona Costera Patagónica. Informe Técnico, no 20
Evans KL, Warren PH, Gaston KJ (2005) Species–energy relationships at the macroecological scale: a review of the mechanisms. Biol Rev Camb Philos Soc 80:1–25
Friedman JH (2002) Stochastic gradient boosting. Comput Stat Data Anal 38:367–378
Gaertner JC, Bertrand JA, Gil de Sola L, Durbec JP, Ferrandis E, Souplet A (2005) Large spatial scale variation of demersal fish assemblage structure on the continental shelf of the NW Mediterranean Sea. Mar Ecol Prog Ser 297:245–257
Gagliardini DA, Rivas AL (2004) Environmental characteristics of San Matías Gulf obtained from LANDSAT-TM and ETM+ data. Gayana (Concepc) 68:186–193
Gagliardini DA, Aliotta S, Dogliotti AI, Clemente-Colón P (2005) Identification of bed forms through ERS SAR images in San Matías Gulf, Argentina. J Coast Res 211: 193–201
Garofalo G, Fiorentino F, Gristina M, Cusumano S, Sinacori G (2007) Stability of spatial pattern of fish species diversity in the Strait of Sicily (central Mediterranean). In: Relini G, Ryland J (eds) Biodiversity in enclosed seas and artificial marine habitats. Developments in hydrobiology, Vol 193. Springer, Dordrecht, p 117–124
Gaston KJ (2000) Global patterns in biodiversity. Nature 405:220–227
Gaston K, Blackburn T (2008) Pattern and process in macroecology. John Wiley & Sons, New York, NY
González R, Gaspar C, Curtolo L, Sangiuliano I, Osovnikar P, Borsetta N (2004) Fishery and oceanographic monitoring system (FOMS): a new technological tool based on remote sensing, with application in ecosystem management of coastal fisheries in Patagonia. Gayana (Concepc) 68:234–238
González R, Ocampo Reinaldo M, Schneider C, Romero MA and others (2010) Correlating SST satellite data to the spatial distribution of spawning aggregations of Argentine hake (*Merluccius hubbsi*) in San Matías Gulf, Patagonia, Argentina. In: Proceedings of Oceans From Space Symposium. Joint Research Centre, European Commission JRC Scientific and Technical Reports, no. 57986. Venice, p 103–104
Gotelli NJ (2008) A primer of ecology, 4th edn. Sinauer, Sunderland, MA
Gotelli NJ, Shimadzu H, Dornelas M, McGill B, Moyes F, Magurran AE (2017) Community-level regulation of temporal trends in biodiversity. Sci Adv 3:e1700315
Guisan A, Zimmermann NE (2000) Predictive habitat distribution models in ecology. Ecol Modell 135:147–186
Hijmans RJ, Phillips S, Leathwick J, Elith J, Hijmans MRJ (2017) Package 'dismo'. Circles 9:1–68
Hurlbert SH (1971) The nonconcept of species diversity: a critique and alternative parameters. Ecology 52:577–586
Isla FI (2013) The flooding of the San Matías Gulf: the Northern Patagonia sea-level curve. Geomorphology 203: 60–65
Jeres P, Romero MA, Svendsen GM, Ocampo-Reinaldo M, González R (2015) Análisis de la composición y distribución de los ensambles ícticos demersales del Golfo San Matías, Patagonia. In: Riccialdelli L, Sotelano MP, Boy CC (eds) Libro de Resúmenes IX Jornadas Nacionales de Ciencias del Mar y XVII Coloquio de Oceanografía: Ciencia y Sociedad: Integrando saberes en los estudios del mar, p 360
Johnson CN, Balmford A, Brook BW, Buettel JC, Galetti M, Guangchun L, Wilmshurst JM (2017) Biodiversity losses and conservation responses in the Anthropocene. Science 356:270–275
Leathwick JR, Elith J, Francis MP, Hastie T, Taylor P (2006) Variation in demersal fish species richness in the oceans surrounding New Zealand: an analysis using boosted regression trees. Mar Ecol Prog Ser 321:267–281
Leathwick J, Moilanen A, Francis M, Elith J and others (2008) Novel methods for the design and evaluation of marine protected areas in offshore waters. Conserv Lett 1:91–102
Lucifora LO, García VB, Menni RC, Worm B (2012) Spatial patterns in the diversity of sharks, rays, and chimaeras (Chondrichthyes) in the Southwest Atlantic. Biodivers Conserv 21:407–419
MacArthur RH, Wilson EO (1967) The theory of island biogeography. Princeton University Press, Princeton, NJ
Magnussen E (2002) Demersal fish assemblages of Faroe Bank: species composition, distribution, biomass spectrum and diversity. Mar Ecol Prog Ser 238:211–225

Author copy

Magurran AE, Khachonpisitsak S, Ahmad AB (2011) Biological diversity of fish communities: pattern and process. J Fish Biol 79:1393–1412

Magurran AE, Dornelas M, Moyes F, Gotelli NJ, McGill B (2015) Rapid biotic homogenization of marine fish assemblages. Nat Commun 6:8405

Mannocci L, Boustany AM, Roberts JJ, Palacios DM and others (2017) Temporal resolutions in species distribution models of highly mobile marine animals: recommendations for ecologists and managers. Divers Distrib 23:1098–1109

Martínez-Rincón RO, Ortega-García S, Vaca-Rodríguez JG (2012) Comparative performance of generalized additive models and boosted regression trees for statistical modeling of incidental catch of wahoo (*Acanthocybium solandri*) in the Mexican tuna purse-seine fishery. Ecol Modell 233: 20–25

Mazio CA, Vara CD (1983) Las mareas del golfo San Matías. Departamento Oceanografía, Servicio de Hidrografía Naval, Informe Técnico, Buenos Aires

McGill BJ (2010) Towards a unification of unified theories of biodiversity. Ecol Lett 13:627–642

Menni RC (1983) Los peces en el medio marino. Estudio Sigma, Buenos Aires

Mittelbach GG, Steiner CF, Scheiner SM, Gross KL and others (2001) What is the observed relationship between species richness and productivity? Ecology 82:2381–2396

Moore CH, Harvey ES, Van Niel K (2010) The application of predicted habitat models to investigate the spatial ecology of demersal fish assemblages. Mar Biol 157:2717–2729

Ocampo Reinaldo M, González R, Romero MA (2011) Feeding strategy and cannibalism of the Argentine hake *Merluccius hubbsi*. J Fish Biol 79:1795–1814

Ocampo Reinaldo M, González R, Williams G, Storero LP, Romero MA, Narvarte M, Gagliardini DA (2013) Spatial patterns of the Argentine hake *Merluccius hubbsi* and oceanographic processes in a semi-enclosed Patagonian ecosystem. Mar Biol Res 9:394–406

Ocampo Reinaldo MO, Milessi AC, Romero MA, Crespo E, Wolff M, González RA (2016) Assessing the effects of demersal fishing and conservation strategies of marine mammals over a Patagonian food web. Ecol Modell 331: 31–43

Olson D, Hitchcock G, Mariano A, Ashjian C, Peng G, Nero R, Podesta G (1994) Life on the Edge: Marine Life and Fronts. Oceanography 7:52–60

Pierce JW, Sieguel FR, Urien CM (1969) Topografía submarina del Golfo San Matías. Proceedings of the VI Jornadas Geológicas Argentinas, p 127–140

Pimm SL, Jenkins CN, Abell R, Brooks TM and others (2014) The biodiversity of species and their rates of extinction, distribution, and protection. Science 344:1246752

Piola AR, Scasso LML (1988) Corrientes en el Golfo San Matías. Geoacta 15:33–51

Pisoni JP, Rivas AL, Piola AR (2015) On the variability of tidal fronts on a macrotidal continental shelf, Northern Patagonia, Argentina. Deep Sea Res II 119:61–68

Platt T, Sathyendranath S (1988) Oceanic primary production: estimation by remote sensing at local and regional scales. Science 241:1613–1620

R Core Development Team (2018) R: a language and environment for statistical computing. R Foundation for Statistical Computing, Vienna

Rapoport EH (1994) Remarks on marine and continental biogeography: an areographical viewpoint. Philos Trans R Soc B 343:71–78

Ridgeway G (2006) Gbm: Generalized boosted regression models. R package version 1:55. https://cran.r-project.org/web/packages/gbm/index.html

Rivas AL, Pisoni JP (2010) Identification, characteristics and seasonal evolution of surface thermal fronts in the Argentinean Continental Shelf. J Mar Syst 79:134–143

Roberts JJ, Best BD, Dunn DC, Treml EA, Halpin PN (2010) Marine Geospatial Ecology Tools: An integrated framework for ecological geoprocessing with ArcGIS, Python, R, MATLAB, and C++. Environ Model Softw 25:1197–1207

Romero MA, Reinaldo MO, Williams G, Narvarte M, Gagliardini DA, González R (2013) Understanding the dynamics of an enclosed trawl demersal fishery in Patagonia (Argentina): a holistic approach combining multiple data sources. Fish Res 140:73–82

Rosenzweig ML (1995) Species diversity in space and time. Cambridge University Press, Cambridge

Smoliński S, Radtke K (2017) Spatial prediction of demersal fish diversity in the Baltic Sea: comparison of machine learning and regression-based techniques. ICES J Mar Sci 74:102–111

Soininen J, Lennon JJ, Hillebrand H (2007) A multivariate analysis of beta diversity across organisms and environments. Ecology 88:2830–2838

Soykan CU, Eguchi T, Kohin S, Dewar H (2014) Prediction of fishing effort distributions using boosted regression trees. Ecol Appl 24:71–83

Sperling EA, Frieder CA, Levin LA (2016) Biodiversity response to natural gradients of multiple stressors on continental margins. Proc R Soc B 283:20160637

Tamdrari H, Brêthes JC, Archambault D (2015) Spatio-temporal variations in demersal fish assemblages and diversity in the northern Gulf of St. Lawrence (Canada). Mar Ecol 36:557–571

Tittensor DP, Mora C, Jetz W, Lotze HK, Ricard D, Berghe EV, Worm B (2010) Global patterns and predictors of marine biodiversity across taxa. Nature 466:1098–1101

Tonini MH, Palma ED, Piola AR (2013) A numerical study of gyres, thermal fronts and seasonal circulation in austral semi-enclosed gulfs. Cont Shelf Res 65:97–110

Vaquer-Sunyer R, Duarte CM (2008) Thresholds of hypoxia for marine biodiversity. Proc Natl Acad Sci USA 105: 15452–15457

Williams GN (2011) Caracterización ambiental del golfo San Matías mediante sensores remotos y parámetros oceanográficos: relación con la distribución y abundancia de los recursos biológicos de interés pesqueros. PhD thesis, Universidad Nacional del Comahue, Bariloche

Williams G, Sapoznik M, Ocampo-Reinaldo M, Solis M and others (2010) Comparison of AVHRR and SeaWiFS imagery with fishing activity and *in situ* data in San Matías Gulf, Argentina. Int J Remote Sens 31:4531–4542

Worm B, Barbier EB, Beaumont N, Duffy JE and others (2006) Impacts of biodiversity loss on ocean ecosystem services. Science 314:787–790

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